1. Digital Quantum Circuits: Pictorial identities

1.1. Symbolic alphabet

These symbols form the "alphabet" of digital quantum circuit design, facilitating communication of operations in a visual and standardized way.[1]

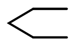
Wires

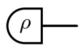
Quantum wire

Quantum wire bundle (n qubits) 

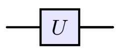
Quantum wire bundle (alternate) 

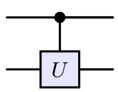
Classical wire

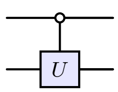
Entangled bit (ebit; Bell pair) 

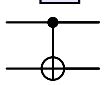
Input quantum state ρ 

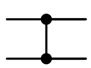
Unitary gates

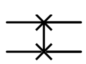
Quantum unitary gate U (same box used for non-unitary quantum channels) 

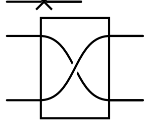
Control gate U (control on $|1\rangle$) 

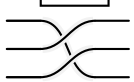
Control gate U (control on $|0\rangle$) 

Control-X (cNOT) 

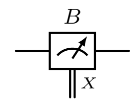
Control-Z (cZ)[2] 

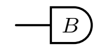
Swap gate 

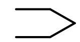
Swap gate (alternate) 

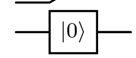
Wire permutation (swap network) 

Measurements

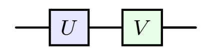
Measurement in basis B with classical outcome X (random variable) 

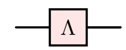
Measurement in basis B: alternative pictogram 

Measurement in the Bell basis 

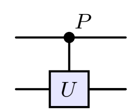
Reset channel 

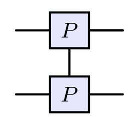
Accents

Primary and secondary color accents for gates 

Noisy gate or channel 

Advanced notation

P-controlled U gate: The control acts on the $|-P\rangle$ eigenstate of Pauli P (e.g., X, Y, Z). For a standard controlled-U gate, $P = Z$, meaning control is on $|-Z\rangle = |1\rangle$. 

P-controlled P gate: An alternative notation highlighting the symmetry between control and target. A key example is $P = U = Z$, resulting in the symmetric cZ gate, where control and target roles are interchangeable. 

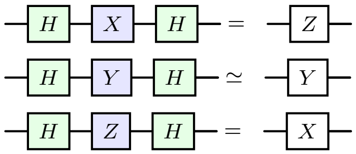
[1] This graphical notion was first made explicit in R. Feynman, "Quantum mechanical computers" (1986) and D. Deutsch, "Quantum computational networks" (1989); see also Barenco, et al. "Elementary gates for quantum computation" (1995). Note that Penrose (tensor network notation) commuted the Deutch paper.

1.2. Paulis: Basic and super useful

Global phases noted below are for unitaries, not channels. Proofs and checks: See (C69) and Mathematica/2021 RC Learn experiment/2022-11.1 circuit identities basic.nb

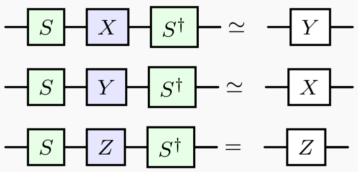
Pauli basis changes (conjugation)

H-**gate conjugation** of XYZ: Changes $X \leftrightarrow Z$ bases.



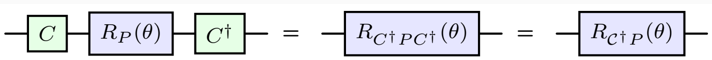
(-1 global phase)

S-**gate conjugation** of XYZ



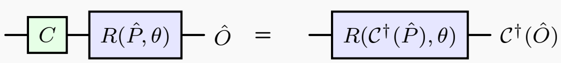
(-1 global phase)

Pauli rotation conjugation

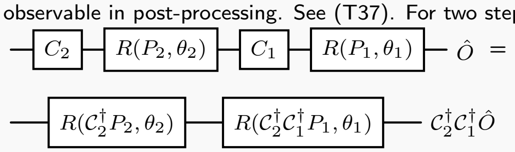


See (T37). For a rotation around a Pauli P and conjugation by Clifford gate C. Note the order of operations.

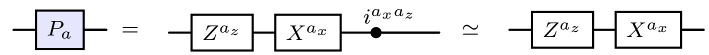
Pauli rotation + Clifford compiling



For final observable $\hat{O}$, we can push the Clifford gate through the Pauli rotation and handle the observable in post-processing. See (T37). For two steps:

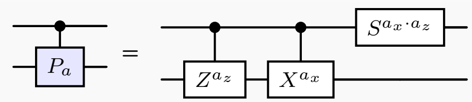


Pauli gate decomposition



Pauli decomposition $P_a = i^{a_x a_z} X^{a_x} Z^{z_z}$ with $a = (a_x, a_z)$, see Sec. **??** and **??**. Note that for gates $P_a \cdot P_a^\dagger$, the $i^{a_x a_z}$ drops out. The global phase $i^{a_x a_z}$ for non control Pauli P_a can be ignored. It only applied to $a = (1\ 1)$ for $P_a = Y$ (C70-4).

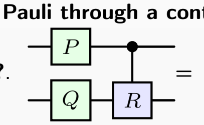
Control-Pauli decomposition



Note that the phase kickback from the Y Pauli P_a shows up as a i phase on the control 1 state. Not this S gate only shows up for Y, i.e., $a = (1\ 1)$ (C70-4).

Pushing a Pauli through a control-Pauli

See Sec. **??**.

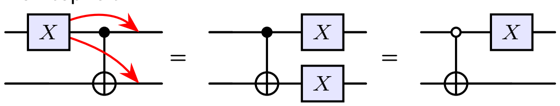


1.3. Controlled-NOT Gate (cNOT, cX) and SWAPs

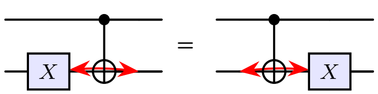
X through cX

***X* travels forwards from control** to target (C69-2)

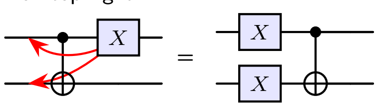
X on top left



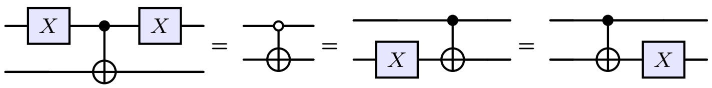
X on bottom left / right



X on top right



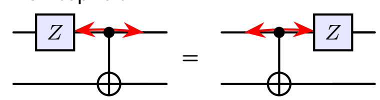
CX conjugated by X on control (on target commutes) (C69-1)



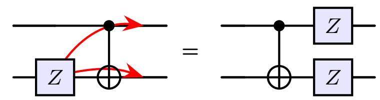
Z through cX

***Z* travels backwards** from target to control (C69-5 / see above)

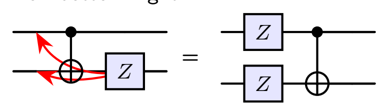
Z on top left



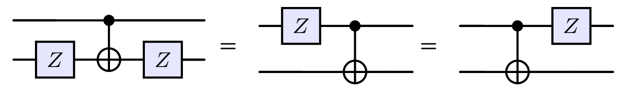
Z on bottom left



Z on bottom right

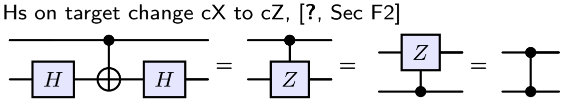


CX conjugated by Z on target (on control commutes) (C69-5)

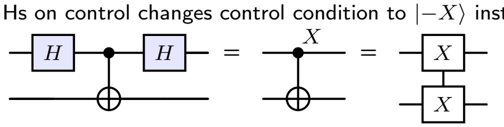


H conjugating cX

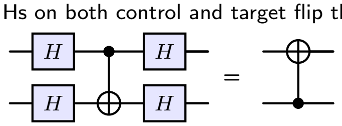
Hs on target change cX to cZ, [?, Sec F2]



Hs on control changes control condition to $|-X\rangle$ instead of $|-Z\rangle$. Last equivalence is just notation.

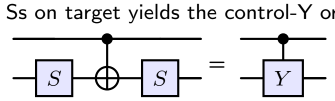


Hs on both control and target flip the CX direction



S conjugating cX

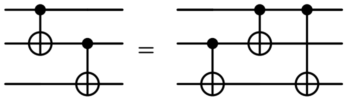
Ss on target yields the control-Y or cY gate [Maslov]



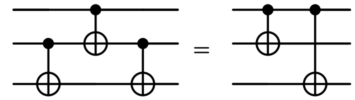
Ss on control yields ...

Ss on control and target yields ....

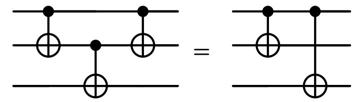
Multiple cNOTs



(C69-3) Note that $\mathrm{cX}_{12}\mathrm{cX}_{13} = |0\rangle\langle 0|_1 I + |1\rangle\langle 1|_1 X_2 X_3$. Think of having to push through the X on the control-2 only when control-1 is $|1\rangle\langle 1|$, then pushing X through cX_{23} yields X travels forwards and gives an X on each of the 2-3 wires.



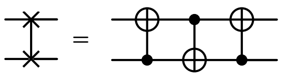
(4.1 of Maslov)



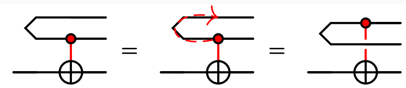
(4.6 of Maslov)

SWAP gates

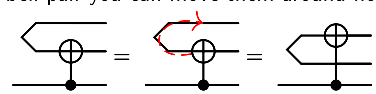
SWAP



SWAP + more wires

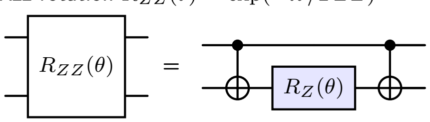


Note that because we have a Bell pair, it doesn't matter which wire the target or the control is of the bell pair you can move them around no problem

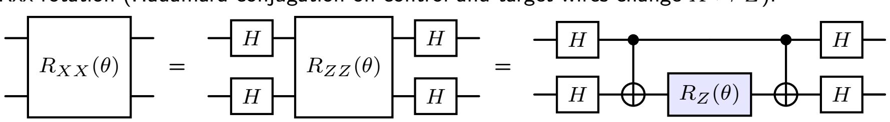


Pauli-rotation simulation gates

Rzz rotation $R_{ZZ}(\theta) = \exp(-i\theta/2\, ZZ)$



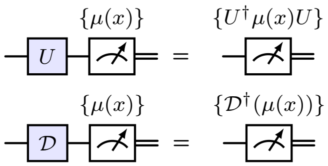
Rxx rotation (Hadamard conjugation on control and target wires change $X \mapsto Z$).



Ryy can be converted to Rzz with conjugation on both (c) and (t) wires by $R_{YY}(\theta) = (SS).(HH).R_{ZZ}(\theta).(HH)^\dagger.(SS)^\dagger$, see the Pauli conjugation identities in Sec.

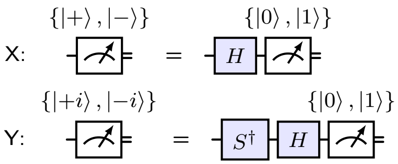
1.4. Measurement based

Rotating/acting on the measurement basis



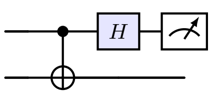
(Sec. **??** and **??**)

X and Y basis measurement



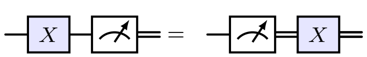
(Sec. **??**)

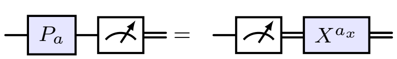
Bell basis measurement



Projects in Bell Basis N&C Ex4.33 (Sec. **??**)

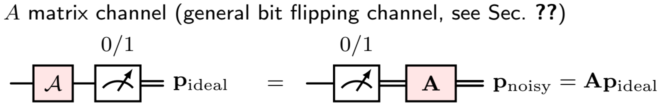
Bit flip quantum to classical





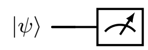
General Pauli P_a

A matrix channel (general bit flipping channel, see Sec. **??**)



Principle of implicit measurement

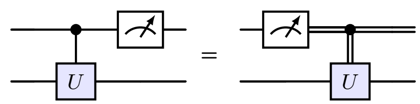
Without loss of generality, any unterminated quantum wires (qubits which are not measured) at the end of a quantum circuit may be assumed to be measured. (Nielsen Chuang pg 187)



Principle of deferred measurement

Measurements can always be moved from an intermediate stage of a quantum circuit to the end of the circuit; if the measurement results are used at any stage of the circuit then the classically controlled operations can be replaced by conditional quantum operations. (Nielsen Chuang pg. 186)
* Note, measurements can always be represented by unitary transforms with ancilla qubits followed by projective measurements.

Measurement commutes with controls



Consequence of the principle of deferred measurement [**?**, Ex 4.35]

TODO: Add here the identities from (T35) on pushing the X gates through the measurement and conditional channel, as well as noise channels and bit flips.

1.4.1. Multi-qubit controls

Three qubit control gate identities ...

- Identity references: D. Maslov